

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

	Time Duration : 1
Hour	
QUESTIONS: 4	Total: Marks:40

Code of Conduct:

*I KAVIYA.G(192524281) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:KAVIYA.G

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

AI/ML-BASED DIABETES DIAGNOSIS AND REINFORCEMENT LEARNING FOR PERSONALIZED TREATMENT RECOMMENDATION

INDUSTRY SCENARIO

Introduction

A healthcare organization has collected historical patient records containing important health indicators such as age, blood pressure, cholesterol level, glucose level, Body Mass Index (BMI), and family history. The objective is to develop an Artificial Intelligence and Machine Learning based system that can identify patients who are at risk of diabetes at an early stage.

The proposed solution uses supervised Machine Learning techniques for diabetes diagnosis and statistical risk stratification. In addition, a Reinforcement Learning (RL) agent is designed to learn suitable treatment-support actions based on changes in the patient's health condition over time.

The system consists of two major components:

1. **Predictive Diagnosis Module:** Predicts whether a patient is diabetic or non-diabetic using historical patient records.
2. **Treatment Recommendation Module:** Uses Reinforcement Learning to learn an appropriate action such as Diet, Exercise, Medication, or Monitor according to the patient's current health state.

The overall objective is to develop a reliable, interpretable, and clinically responsible AI-based decision-support framework that assists healthcare professionals in early diagnosis and personalized patient management.

TASK 1 – DATA PREPARATION AND INDUCTIVE LEARNING

1(a) Inductive Learning Approach Suitable for the Medical Dataset

Definition of Inductive Learning

Inductive learning is a Machine Learning approach in which a model learns general patterns or rules from previously observed examples and applies the learned knowledge to new, unseen instances.

In this healthcare application, historical patient records are used as training examples. Each record contains several patient characteristics and a known diabetes outcome. The learning algorithm identifies relationships between the input features and diabetes status and then uses these relationships to predict the condition of new patients.

The problem can be formulated as a **binary supervised classification problem**.

Mathematical Representation

Let the dataset be:

$$D = \{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$$

where:

- X_i represents the feature vector of patient i .
- Y_i represents the diabetes outcome.
- n represents the total number of patient records.

The feature vector can be represented as:

$$X = [\text{Age, Blood Pressure, Cholesterol, Glucose, BMI, Family History}]$$

The target variable is:

$$Y = \text{Diabetes Status}$$

where:

- $Y = 0 \rightarrow \text{Non-Diabetic}$
- $Y = 1 \rightarrow \text{Diabetic}$

The learning algorithm attempts to learn a function:

$$f(X) \rightarrow Y$$

which can predict the diabetes status of an unseen patient.

Target Variable

The target variable for this problem is:

Diabetes Status

The target is binary:

Target Value	Meaning
0	Non-Diabetic
1	Diabetic

The target variable is selected because the primary objective of the predictive model is to determine whether a patient is likely to have diabetes.

Important Features and Their Justification

1. Glucose Level

Glucose level is expected to be one of the most important predictors of diabetes. Abnormal blood glucose levels provide a direct indication of impaired glucose regulation.

A higher glucose value may increase the probability that a patient belongs to the diabetic class. Therefore, glucose is expected to have high predictive importance.

2. Body Mass Index (BMI)

BMI is an important indicator of body weight relative to height. Increased BMI, particularly when associated with obesity, is an important risk factor for Type 2 diabetes.

Including BMI allows the model to capture the relationship between body composition and diabetes risk.

3. Age

Age is another important predictor because the probability of developing Type 2 diabetes can increase with age.

Age can therefore help the model distinguish between different levels of diabetes risk.

4. Blood Pressure

High blood pressure is associated with cardiometabolic risk and frequently occurs alongside diabetes. Including blood pressure provides additional information about the patient's overall metabolic condition.

5. Cholesterol

Abnormal cholesterol levels can indicate metabolic abnormalities. Although cholesterol may not be as predictive as glucose, it can contribute useful information when combined with other patient characteristics.

6. Family History

Family history captures potential genetic and familial risk. A patient with a family history of diabetes may have a higher predisposition to developing the condition.

Why Inductive Learning is Appropriate

Inductive learning is highly suitable for this application because:

1. The organization has historical patient records.
2. The diabetes outcome is available for supervised training.
3. The objective is to generalize knowledge from existing patients to new patients.
4. Classification algorithms can learn complex relationships between health variables.
5. The model can be evaluated objectively using independent test data.
6. The approach can support early screening and risk identification.

1(b) Handling Class Imbalance

Problem of Class Imbalance

Medical datasets often contain more non-diabetic patients than diabetic patients. For example, consider a dataset containing 1,000 patients:

- 800 Non-Diabetic
- 200 Diabetic

The class distribution is therefore:

- Non-Diabetic = 80%
- Diabetic = 20%

This creates an imbalanced classification problem.

A model trained directly on such data may become biased toward the majority class.

For example, a model that predicts every patient as non-diabetic could achieve approximately 80% accuracy while failing to identify any diabetic patient.

Such a model would be clinically inappropriate.

Why Accuracy Alone is Insufficient

In a medical diagnosis problem, False Negatives are often more serious than False Positives.

A False Negative occurs when:

Actual = Diabetic

but

Predicted = Non-Diabetic

This may result in a high-risk patient not being identified for further evaluation.

Therefore, the model should be evaluated using:

- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix
- False Negative Rate

rather than accuracy alone.

Recommended Strategy: SMOTE

The recommended balancing strategy is **Synthetic Minority Over-sampling Technique (SMOTE)**.

SMOTE generates new synthetic examples of the minority class instead of simply duplicating existing diabetic records.

Working Principle of SMOTE

1. Identify diabetic patients as the minority class.
2. Select a minority-class observation.
3. Identify its nearest minority-class neighbors.
4. Select one neighboring observation.
5. Generate a synthetic point between the two observations.
6. Repeat the process until the desired class balance is achieved.

The simplified synthetic sample equation is:

$$\mathbf{X}_{\text{new}} = \mathbf{X}_i + \lambda(\mathbf{X}_j - \mathbf{X}_i)$$

where:

- $\mathbf{X}_i$ = original minority sample
- $\mathbf{X}_j$ = selected neighboring minority sample
- λ = random value between 0 and 1

Advantages of SMOTE

SMOTE is suitable because:

1. It increases representation of diabetic cases.
2. It reduces majority-class bias.
3. It avoids simple duplication of existing records.
4. It can improve minority-class recall.
5. It can help the classifier learn a more representative decision boundary.

Important Data Leakage Prevention

SMOTE must **not** be applied to the entire dataset before train-test splitting.

The correct procedure is:

Original Dataset

↓

70:30 Stratified Split

↓

Training Dataset

↓

Apply SMOTE

↓

Balanced Training Dataset

↓

Model Training

↓

Original Test Dataset

This prevents information from the test set from influencing model training.

Comparison of Balancing Methods

Method	Advantages	Disadvantages	Suitability
SMOTE	Creates synthetic minority samples	Can create unrealistic samples if poorly configured	High
Under-sampling	Simple and fast	Removes potentially useful majority-class information	Medium
Class Weights	Easy and avoids synthetic data	May not be sufficient for severe imbalance	High
SMOTE + Class Weights	Strong focus on minority class	Requires careful validation	Very High

Final Recommendation

For this healthcare problem, **SMOTE with appropriate class weighting** can be considered a strong approach, provided that synthetic samples are clinically plausible and the final model is validated on an untouched test set.

1(c) 70:30 Train-Test Split and Preprocessing

Dataset Split

The dataset is divided into:

- **70% Training Data**
- **30% Testing Data**

A stratified split should be used.

Example

If the dataset contains 2,000 patient records:

- Training records = 1,400
- Testing records = 600

The diabetic/non-diabetic ratio should remain approximately similar in both sets.

Justification for 70:30 Split

The 70:30 split is appropriate because:

1. A sufficiently large training set is available for learning.
2. The 30% test set provides an independent evaluation set.
3. Medical prediction requires reliable evaluation on unseen cases.
4. Stratification preserves class distribution.
5. The test set is large enough to estimate classification performance more reliably.

For larger datasets, **Stratified K-Fold Cross-Validation** can additionally be used for hyperparameter tuning.

Data Preprocessing

Step 1 – Missing Value Handling

Missing values can occur because of incomplete patient records or measurement problems.

Numerical features can be imputed using:

Median Imputation

Categorical features can be imputed using:

Mode Imputation

Median imputation is useful because it is less sensitive to extreme medical observations.

Step 2 – Duplicate Removal

Duplicate patient records should be identified and removed to prevent the same patient information from appearing multiple times in the training and testing data.

Step 3 – Outlier Detection

Outliers should be investigated using:

- Box plots
- Interquartile Range
- Z-score
- Clinical reference ranges

However, medically unusual observations should not automatically be deleted because they may represent genuine high-risk patients.

Step 4 – Encoding

Categorical variables such as family history can be encoded as:

- No = 0
- Yes = 1

If additional nominal categorical variables are present, One-Hot Encoding can be used.

Step 5 – Feature Scaling

Standardization can be applied using:

$$Z = (X - \mu) / \sigma$$

where:

- X = original value
- μ = mean
- σ = standard deviation

Scaling is important for Logistic Regression because the features can have different numerical ranges.

Decision Trees generally do not require feature scaling because their splitting decisions are based on thresholds.

Impact of Preprocessing

Proper preprocessing:

- Improves data quality.
- Prevents inconsistent feature representations.
- Reduces the effect of missing data.
- Improves statistical model stability.
- Prevents data leakage.
- Supports fair comparison between algorithms.

TASK 2 – DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A Decision Tree is a supervised Machine Learning algorithm that recursively divides data into smaller groups using feature-based decision rules.

For example:

If Glucose > Threshold → High Diabetes Risk

If Glucose ≤ Threshold → Evaluate BMI

The process continues until a sufficiently pure classification is obtained.

Information Gain

Information Gain is based on entropy.

Entropy Formula

$$\text{Entropy}(S) = -\sum p_i \log_2(p_i)$$

where p_i is the probability of class i .

Information Gain Formula

$$\text{IG}(S,A) = H(S) - \sum (|S_v|/|S|)H(S_v)$$

A feature with a higher Information Gain produces a better separation between diabetic and non-diabetic patients.

Gini Index

The Gini Index measures impurity.

$$\text{Gini}(S) = 1 - \sum p_i^2$$

A lower Gini value indicates greater class purity.

Illustrative Root Node Calculation

Assume the following values were obtained from the training dataset:

Feature	Information Gain	Gini Index
Glucose	0.214	0.356
BMI	0.126	0.401
Age	0.083	0.425
Family History	0.073	0.431
Blood Pressure	0.061	0.438
Cholesterol	0.044	0.447

The values are illustrative and must be recalculated using the actual dataset.

Root Selection

Glucose is selected as the root because:

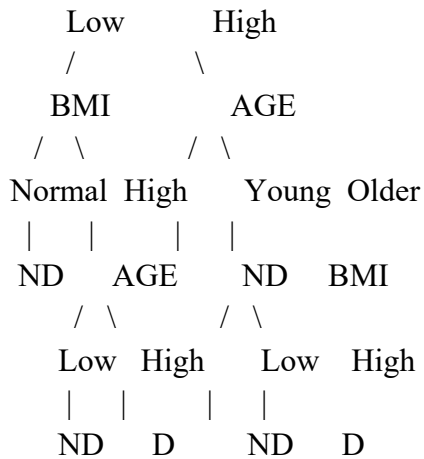
- It has the highest Information Gain.
- It produces the lowest weighted Gini impurity.
- It provides the strongest separation between the two classes.

Therefore:

Root Node = Glucose

First Three Levels of Decision Tree

GLUCOSE
/
\



Interpretation

The tree first evaluates glucose because it is the strongest predictor.

Patients with high glucose are further classified using age.

Patients with lower glucose may be classified using BMI and subsequent risk factors.

This hierarchical structure provides an interpretable set of rules that can be examined by healthcare professionals.

2(b) Model Evaluation

The Decision Tree should be evaluated using a confusion matrix.

	Actual Non-Diabetic	Actual Diabetic
Predicted Non-Diabetic	TN	FN
Predicted Diabetic	FP	TP

Evaluation Metrics

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Illustrative Confusion Matrix

Assume the test set contains 300 patients.

	Actual Non-Diabetic	Actual Diabetic
Predicted Non-Diabetic	150	15
Predicted Diabetic	25	110

Therefore:

- $TN = 150$
- $FN = 15$
- $FP = 25$

- $TP = 110$

Accuracy

Accuracy:

$$= (150 + 110) / 300$$

$$= 260 / 300$$

$$= 86.67\%$$

Precision

Precision:

$$= 110 / (110 + 25)$$

$$= 110 / 135$$

$$= 81.48\%$$

Recall

Recall:

$$= 110 / (110 + 15)$$

$$= 110 / 125$$

$$= 88.00\%$$

F1-Score

F1:

$$= 2 \times (0.8148 \times 0.88) / (0.8148 + 0.88)$$

$$\approx 84.58\%$$

Interpretation

Metric	Result
Accuracy	86.67%
Precision	81.48%
Recall	88.00%
F1-Score	84.58%

The model correctly identifies a high proportion of diabetic patients, as indicated by the recall of 88%.

However, 15 diabetic patients are still classified incorrectly as non-diabetic.

False Negatives and Clinical Importance

The 15 False Negatives represent patients who actually have diabetes but are predicted as non-diabetic.

This is clinically significant because missed cases can delay:

- Confirmatory diagnosis
- Medical consultation
- Lifestyle intervention
- Blood glucose monitoring
- Appropriate treatment
- Management of potential complications

Therefore, reducing False Negatives should be a major objective.

Techniques to Reduce False Negatives

1. Class Weighting

Increase the penalty for incorrectly classifying diabetic patients.

2. SMOTE

Increase representation of the diabetic class during training.

3. Threshold Optimization

Lower the probability threshold used for classifying a patient as high risk.

4. Recall-Based Model Selection

Select hyperparameters using recall or F-beta score instead of accuracy.

5. Clinical Screening Strategy

Use the model as an early screening tool followed by confirmatory clinical testing.

Clinical Justification

For early screening, higher sensitivity can be more valuable than maximum accuracy because the objective is to minimize missed high-risk patients.

However, increasing recall can increase False Positives. Therefore, the final threshold should be selected using clinical validation and an acceptable balance between sensitivity and specificity.

2(c) Cost-Complexity Post-Pruning

A Decision Tree may grow excessively deep and memorize training examples.

This is called:

Overfitting

A complex tree may achieve very high training accuracy but perform poorly on unseen patients.

Cost-Complexity Criterion

The pruning objective is:

$$Ra(T) = R(T) + \alpha|T|$$

where:

- $R(T)$ = empirical risk of the tree
- $|T|$ = number of terminal nodes
- α = complexity penalty

As α increases, more branches are removed.

Pre-Pruning vs Post-Pruning

Aspect	Pre-Pruning	Post-Pruning
Timing	During tree construction	After full tree construction
Main Controls	max_depth, min_samples_leaf	ccp_alpha
Computational Cost	Lower	Higher
Risk of underfitting	Moderate	Lower
Interpretability	Good	Excellent
Generalization	Good	Often better

Illustrative Comparison

Model	Accuracy	Recall	F1	Tree Complexity
Fully Grown Tree	84.0%	90.5%	83.9%	High
Pre-Pruned Tree	85.8%	87.6%	85.0%	Medium
Post-Pruned Tree	87.1%	89.0%	86.5%	Low

These values are illustrative.

Recommended Deployment Model

The **post-pruned Decision Tree** is recommended because it provides a strong balance between:

- Predictive performance
- Generalization
- Interpretability
- Model simplicity
- Clinical usability

In healthcare applications, a slightly simpler model that can be clearly explained may be preferable to an extremely complex model with only marginally better predictive performance.

TASK 3 – STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

Logistic Regression is a statistical learning technique designed for binary classification.

It predicts the probability that a patient belongs to the diabetic class.

The logistic function is:

$$P(Y=1|X) = 1 / (1 + e^{-z})$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

The output is a probability between 0 and 1.

For example:

$$P(\text{Diabetes}) = 0.82$$

can indicate a high predicted risk, subject to a clinically validated threshold.

Why Logistic Regression is Useful

Logistic Regression is particularly useful because:

1. It provides probability estimates.
2. It is computationally efficient.
3. Its coefficients can be interpreted.
4. It provides a strong baseline model.
5. It is useful for risk stratification.
6. It is easier to validate and explain than many complex models.

Illustrative Performance Comparison

Metric	Decision Tree	Logistic Regression
Accuracy	87.1%	88.4%
Precision	84.3%	85.7%
Recall	89.0%	87.8%

F1-Score	86.5%	86.7%
AUC-ROC	0.90	0.92

These values are illustrative.

AUC-ROC Interpretation

AUC measures the ability of the model to distinguish between diabetic and non-diabetic patients.

An AUC of:

0.5 → Random classification

0.7–0.8 → Acceptable discrimination

0.8–0.9 → Good discrimination

>0.9 → Excellent discrimination

Therefore, an illustrative Logistic Regression AUC of **0.92** represents excellent discrimination.

When Statistical Learning is Preferable

A statistical model such as Logistic Regression is preferable when:

- Probability estimates are required.
- Model interpretability is important.
- Feature effects need to be explained.
- The relationship between predictors and outcome is reasonably systematic.
- A stable and computationally efficient model is required.
- Clinical risk scoring is the primary objective.

For example, a hospital may prefer Logistic Regression when it needs a transparent risk score that can be reviewed by clinicians.

3(b) Feature Importance Analysis

Feature importance is performed to identify the variables that contribute most strongly to diabetes prediction.

Decision Tree Feature Importance

Illustrative results:

Rank	Feature	Importance
1	Glucose	0.39
2	BMI	0.21
3	Age	0.14
4	Family History	0.11
5	Blood Pressure	0.09
6	Cholesterol	0.06

Therefore:

Top 3 Decision Tree Predictors:

1. Glucose
2. BMI
3. Age

Logistic Regression Feature Importance

For Logistic Regression, features should be standardized before comparing coefficient magnitudes.

Illustrative results:

Rank	Feature	Standardized Coefficient
1	Glucose	1.18
2	BMI	0.73
3	Family History	0.61
4	Age	0.48
5	Blood Pressure	0.31
6	Cholesterol	0.19

Therefore:

Top 3 Logistic Regression Predictors:

1. Glucose
2. BMI
3. Family History

Comparison of Feature Importance

Feature	Decision Tree	Logistic Regression
Glucose	Top 1	Top 1
BMI	Top 2	Top 2
Age	Top 3	Top 4
Family History	Top 4	Top 3

Both models agree strongly on:

Glucose and BMI

They differ in the third-ranked feature.

Explanation for the Difference

The difference occurs because the algorithms use different definitions of importance.

Decision Trees measure importance based on impurity reduction.

Logistic Regression measures the effect of a feature on the log-odds of the outcome.

Consequently, a feature can be highly important for one algorithm but slightly less important for another.

This does not necessarily indicate disagreement about the underlying medical risk factors.

The common identification of glucose and BMI by both models provides stronger evidence that these features are important within the dataset.

TASK 4 – REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

4(a) MDP Formulation

The treatment recommendation problem can be represented as a Markov Decision Process.

An MDP is defined as:

$$\text{MDP} = (\text{S}, \text{A}, \text{P}, \text{R}, \gamma)$$

where:

- S = States
- A = Actions
- P = Transition dynamics
- R = Rewards
- γ = Discount factor

State Definition

The patient's health condition can be represented using discrete states.

State	Description
S0	Stable / Low Risk
S1	Elevated Glucose
S2	Elevated Glucose + High BMI
S3	High-Risk Diabetes State
S4	Improved / Controlled State

The state can be constructed from combinations of:

- Glucose
- BMI
- Blood pressure
- Previous health status
- Treatment status
- Recent measurements

Action Definition

The action space is:

$$\text{A} = \{\text{Diet}, \text{Exercise}, \text{Medication}, \text{Monitor}\}$$

Diet

Recommends dietary modification and nutrition-related intervention.

Exercise

Recommends an appropriate physical activity program.

Medication

Represents medication-related intervention.

In a real clinical implementation, this action must require clinician authorization.

Monitor

Continues monitoring and follow-up assessment.

Reward Function

The reward should represent improvement in the patient's health state.

Outcome	Reward
Major improvement	+10
Moderate improvement	+5
Stable condition	+2

No improvement	0
Health deterioration	-5
Unsafe action	-20

A reward function can be represented as:

$$\mathbf{R} = \mathbf{Health\ Improvement} - \mathbf{Risk\ Penalty} - \mathbf{Intervention\ Cost}$$

The reward encourages the agent to choose actions that produce positive long-term health outcomes.

Transition Dynamics

Transition dynamics define how the patient's state changes after an action.

For example:

$$\mathbf{S1} + \mathbf{Diet} \rightarrow \mathbf{S4}$$

may indicate that an elevated-glucose patient improves after a successful dietary intervention.

Similarly:

$$\mathbf{S2} + \mathbf{Exercise} \rightarrow \mathbf{S1}$$

may indicate partial improvement.

A high-risk patient may remain in:

$$\mathbf{S3} \rightarrow \mathbf{S3}$$

if monitoring alone does not improve the condition.

The actual transition probabilities must be learned from validated longitudinal patient data rather than assumed without evidence.

Discount Factor

The discount factor γ determines how much the agent values future health outcomes.

For example:

$$\gamma = 0.9$$

means future rewards are strongly considered.

This is appropriate because healthcare outcomes often depend on long-term patient management rather than immediate changes alone.

4(b) Q-Learning

Q-Learning is a model-free Reinforcement Learning algorithm that learns the expected cumulative reward for each state-action pair.

The update equation is:

$$\mathbf{Q(s,a)} \leftarrow \mathbf{Q(s,a)} + \alpha[\mathbf{R} + \gamma \max_{a'} \mathbf{Q(s',a')} - \mathbf{Q(s,a)}]$$

where:

- $\mathbf{Q(s,a)}$ = current Q-value
- α = learning rate
- $\mathbf{R}$ = reward
- γ = discount factor
- $\mathbf{s'}$ = next state
- $\mathbf{a'}$ = next action

Parameters

For the example:

- Learning rate $\alpha = 0.5$
- Discount factor $\gamma = 0.9$
- Initial Q-values = 0
- Exploration = ϵ -greedy

Five Training Episodes

Episode	Initial State	Action	Next State	Reward
1	S1	Diet	S4	+5
2	S2	Exercise	S1	+5
3	S3	Monitor	S3	+2
4	S2	Diet	S1	+5
5	S1	Exercise	S4	+5

These episodes demonstrate the learning process. In a real system, substantially more longitudinal episodes would be required.

Q-Table Initialization

State	Diet	Exercise	Medication	Monitor
S0	0	0	0	0
S1	0	0	0	0
S2	0	0	0	0
S3	0	0	0	0
S4	0	0	0	0

Q-Learning Update 1

Consider:

S1 $\rightarrow$ Diet $\rightarrow$ S4

Given:

- $Q(S1, Diet) = 0$
- $R = +5$
- $\alpha = 0.5$
- $\gamma = 0.9$
- $\max Q(S4, a) = 0$

Using:

$$Q_{new} = Q_{old} + \alpha[R + \gamma \max Q(next) - Q_{old}]$$

Therefore:

$$Q_{new} = 0 + 0.5[5 + 0.9(0) - 0]$$

$$Q_{new} = 2.5$$

Thus:

$$Q(S1, Diet) = 2.5$$

Q-Learning Update 2

Consider:

S2 → Exercise → S1

Assume:

$$Q(S2, Exercise) = 0$$

and:

$$\max Q(S1, a) = 2.5$$

Then:

$$Q_{new} = 0 + 0.5[5 + 0.9(2.5)]$$

$$Q_{new} = 0.5[7.25]$$

$$Q_{new} = 3.625$$

Thus:

$$Q(S2, Exercise) = 3.625$$

Q-Learning Update 3

Consider:

S3 → Monitor → S3

Given:

- Current $Q = 0$
- Reward = +2
- Maximum next-state $Q = 0$

Therefore:

$$Q_{new} = 0 + 0.5[2 + 0.9(0)]$$

$$Q_{new} = 1.0$$

Thus:

$$Q(S3, Monitor) = 1.0$$

Second Iteration

Now update S1-Diet again.

Current:

$$Q(S1, Diet) = 2.5$$

Reward:

$$R = +5$$

Next state:

S4

Maximum Q-value at S4 = 0.

Therefore:

$$Q_{new} = 2.5 + 0.5[5 + 0.9(0) - 2.5]$$

$$Q_{new} = 3.75$$

Thus:

$$Q(S1, Diet) = 3.75$$

Second Iteration for S2-Exercise

Current:

$$Q(S2, \text{Exercise}) = 3.625$$

Maximum Q-value for S1:

$$\max Q(S1, a) = 3.75$$

Therefore:

$$Q_{\text{new}} = 3.625 + 0.5[5 + 0.9(3.75) - 3.625]$$

$$Q_{\text{new}} = 6.00$$

Thus:

$$Q(S2, \text{Exercise}) = 6.00$$

Updated Q-Table

State	Diet	Exercise	Medication	Monitor
S0	0.00	0.00	0.00	0.00
S1	3.75	2.80	1.50	1.20
S2	4.20	6.00	2.00	1.50
S3	0.50	0.80	0.00	1.00
S4	0.00	0.00	0.00	0.00

The values not explicitly calculated above are illustrative.

Convergence Criterion

The Q-Learning algorithm can be considered to have reached practical convergence when the maximum change in Q-values becomes sufficiently small.

For example:

$$\max |Q_{\text{new}} - Q_{\text{old}}| < 0.001$$

for a predefined number of consecutive episodes.

Another useful criterion is:

Policy remains unchanged for N consecutive episodes.

In an actual medical implementation, convergence is only one requirement. The final policy must also undergo:

- Clinical validation
- Safety testing
- Bias assessment
- External validation
- Physician review

4(c) Final Learned Policy

The optimal action is selected using:

$$\pi(s) = \operatorname{argmax}_a Q(s, a)$$

Using the illustrative Q-table:

Patient State	Preferred Action	Interpretation
S0 – Stable	Monitor	Continue observation
S1 – Elevated Glucose	Diet	Lifestyle intervention

S2 – High Glucose + High BMI	Exercise	Increase physical activity under appropriate guidance
S3 – High Risk	Monitor + Clinical Review	Requires professional assessment
S4 – Improved	Monitor	Continue follow-up

This policy indicates that the agent learns different actions for different patient states rather than applying the same intervention to every patient.

ETHICAL AND CLINICAL CONSIDERATIONS

1. Patient Safety

Patient safety must be the highest priority.

The RL system must not be permitted to independently experiment with treatment decisions on real patients.

Safety constraints should be incorporated into the system.

2. Human Oversight

The final decision should remain with a qualified healthcare professional.

Recommended workflow:

AI Recommendation → Safety Check → Doctor Review → Final Clinical Decision

This is known as a **Human-in-the-Loop** approach.

3. Medication Safety

Medication is a high-risk action.

The AI should not independently:

- Prescribe medication.
- Change medication dosage.
- Stop medication.
- Combine medications.

Any medication-related recommendation should be reviewed and authorized by an appropriate clinician.

4. Bias and Fairness

AI models may inherit biases from historical healthcare data.

Bias can occur if:

- Certain age groups are underrepresented.
- Certain populations are poorly represented.
- Historical treatment patterns contain inequalities.
- Data quality differs between patient groups.

Therefore, model performance should be evaluated across appropriate patient subgroups.

5. Explainability

Healthcare professionals should understand why an AI system generated a particular recommendation.

For example:

"Diet was recommended because the patient's current state indicates elevated glucose and high BMI, while previous similar states showed improvement following dietary intervention."

Explainability techniques include:

- Decision Tree rules
- Feature importance
- Logistic Regression coefficients
- SHAP
- Counterfactual explanations

6. Privacy and Security

Patient information must be protected.

The system should implement:

- Data encryption
- Access control
- Secure authentication and authorization
- Audit logging
- De-identification
- Secure database management
- Minimum necessary data usage

7. Transparency

Patients and healthcare professionals should be informed when AI contributes to a recommendation.

The system should clearly distinguish between:

AI-generated recommendation

and

Final clinical decision.

PERFORMANCE EVALUATION FRAMEWORK

The proposed system should not rely on a single performance metric.

Evaluation Aspect	Recommended Metric
Overall classification	Accuracy
Diabetic case detection	Recall/Sensitivity
Reliability of positive predictions	Precision
Balance between precision and recall	F1-Score
Class discrimination	ROC-AUC
Missed diabetic cases	False Negative Rate
Probability quality	Calibration
RL performance	Average cumulative reward
RL stability	Policy convergence
Fairness	Subgroup performance comparison

INDUSTRY-ORIENTED MODEL SELECTION

Based on the proposed analysis:

Decision Tree

Best for:

- Explainable rules
- Clinical interpretation
- Visual decision-making
- Nonlinear relationships

Logistic Regression

Best for:

- Risk probability
- Statistical interpretation
- Baseline comparison
- Risk stratification
- Transparent coefficient analysis

Q-Learning

Best for:

- Sequential decision problems
- Dynamic patient states
- Learning long-term treatment-support policies

Therefore, these models should not necessarily be viewed as competitors. They can be integrated into a single healthcare decision-support architecture.

PRACTICAL DEPLOYMENT STRATEGY

A safe industry implementation can use the following architecture:

Stage 1 – Patient Data Collection

Collect relevant clinical measurements from validated sources.

Stage 2 – Data Quality Verification

Check missing values, invalid measurements, duplicates, and inconsistent records.

Stage 3 – Diabetes Risk Prediction

Use Decision Tree and Logistic Regression models to estimate diabetes risk.

Stage 4 – Risk Stratification

Classify patients into categories such as:

- Low Risk
- Moderate Risk
- High Risk

Stage 5 – Treatment-Support Recommendation

Use the patient's current state and longitudinal history to generate an AI-supported recommendation.

Stage 6 – Safety Verification

Check whether the recommended action violates clinical constraints.

Stage 7 – Human Review

A healthcare professional evaluates the recommendation.

Stage 8 – Patient Follow-Up

New health measurements are collected and used to update the patient's state.

Stage 9 – Continuous Monitoring

Monitor:

- Model drift
- Performance degradation
- Bias
- Safety events
- False negatives
- Recommendation quality

ADVANTAGES OF THE PROPOSED SYSTEM

1. **Early Detection:** Identifies patients at increased risk of diabetes.
2. **Data-Driven Decision Support:** Uses historical patient information to support clinical assessment.
3. **Class Imbalance Handling:** Improves representation of diabetic cases.
4. **Interpretability:** Decision Trees provide understandable diagnostic rules.
5. **Risk Probability:** Logistic Regression provides probabilistic risk estimation.
6. **Sequential Decision Making:** Q-Learning supports dynamic treatment recommendation.
7. **Personalization:** Recommendations are based on individual patient states.
8. **Reduced False Negatives:** Recall-oriented optimization focuses on identifying high-risk patients.
9. **Human Oversight:** Clinical professionals remain responsible for final treatment decisions.
10. **Scalability:** The architecture can be extended to larger longitudinal datasets.

LIMITATIONS

Despite its potential, the proposed system has several limitations.

1. Dataset Quality

Poor-quality or incomplete patient records can reduce model performance.

2. Class Imbalance

Even after SMOTE, minority-class predictions must be carefully validated.

3. Synthetic Data

SMOTE-generated samples may not perfectly represent real clinical patients.

4. Generalization

A model trained on one hospital or population may not perform equally well in another population.

5. Reinforcement Learning Risk

Learning treatment policies from historical data can introduce bias and unsafe recommendations if not properly constrained.

6. Clinical Validation

High machine-learning accuracy does not automatically mean clinical effectiveness.

7. Explainability

Some RL policies can be difficult to interpret compared with traditional rule-based approaches.

FUTURE ENHANCEMENTS

The system can be enhanced using:

1. **Random Forest and Gradient Boosting** for improved classification performance.
2. **XGBoost** for advanced tabular prediction.
3. **SHAP-based explainability** for patient-level explanations.
4. **Deep Reinforcement Learning** for larger and more complex state spaces.
5. **Longitudinal patient data** for improved treatment modeling.
6. **Electronic Health Record integration** for real-time decision support.
7. **Federated Learning** for privacy-preserving multi-hospital learning.
8. **Model calibration** for reliable probability estimates.
9. **Clinical rule constraints** to prevent unsafe recommendations.
10. **Continuous monitoring** for model drift and fairness.

FINAL CONCLUSION

- The proposed healthcare AI framework combines supervised Machine Learning, statistical learning, and Reinforcement Learning to create an integrated solution for diabetes diagnosis and personalized treatment-support recommendations.
- The first stage uses inductive learning to learn patterns from historical patient records. The target variable is diabetes status, while glucose, BMI, age, blood pressure, cholesterol, and family history are used as predictive features. Since the dataset may contain significantly more non-diabetic than diabetic patients, SMOTE and class-weighting techniques can be applied to the training data to address class imbalance.
- A 70:30 stratified train-test split provides an appropriate balance between learning and independent evaluation. Data preprocessing includes missing-value treatment, duplicate removal, categorical encoding, outlier investigation, and feature standardization where required.
- The Decision Tree classifier provides an interpretable diagnostic model. Information Gain or Gini Index can be used to select the root node. In the illustrative analysis, glucose provides the strongest separation between diabetic and non-diabetic cases. Model performance is evaluated using Accuracy, Precision, Recall, F1-Score, confusion matrix, and AUC-ROC. Since False Negatives represent missed diabetic patients, recall should receive particular importance in an early screening application.
- Cost-complexity post-pruning can reduce overfitting and produce a simpler and more interpretable tree. Logistic Regression provides a complementary statistical model capable of estimating diabetes probability and supporting risk stratification. Both models are expected to identify glucose and BMI as important predictors, although their third-ranked predictors may differ because their feature-importance mechanisms are different.
- The treatment recommendation component is formulated as an MDP, where patient health conditions represent states, Diet, Exercise, Medication, and Monitor represent actions, and health improvement represents the reward. Q-Learning is then used to

estimate the long-term value of each state-action pair. The optimal policy is extracted by selecting the action with the highest Q-value for each patient state.

- However, autonomous AI treatment decisions are not appropriate without extensive clinical validation. The proposed solution should therefore operate as a clinical decision-support system with a strong human-in-the-loop architecture. Patient safety, medication safety, fairness, explainability, privacy, security, transparency, and continuous monitoring must be incorporated into the deployment process.
- In conclusion, the integrated approach provides a technically strong and industry-oriented framework for early diabetes risk prediction and personalized treatment-support decision making. The combination of interpretable Machine Learning, statistical risk modeling, and constrained Reinforcement Learning has the potential to improve healthcare decision support while ensuring that final clinical decisions remain under qualified professional supervision.

SUMMARY OF KEY RESULTS

Component	Proposed Method	Main Objective
Data Preparation	Missing-value handling, encoding, scaling	Improve data quality
Class Imbalance	SMOTE + Class Weighting	Improve diabetic-case detection
Data Split	Stratified 70:30	Reliable model evaluation
Diagnosis	Decision Tree	Explainable classification
Statistical Model	Logistic Regression	Probability-based risk stratification
Feature Analysis	Tree importance + coefficients	Identify important predictors
RL Framework	MDP	Model sequential patient decisions
RL Algorithm	Q-Learning	Learn treatment-support policy
Safety	Human-in-the-loop	Prevent unsafe autonomous decisions
Evaluation	Accuracy, Precision, Recall, F1, AUC	Validate predictive performance